Causal Inference Seminar

Arkādijs Zvaigzne
Econometrics
Stockholm School of Economics in Riga

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Roadmap

Introduction

Omitted Variable Bias

Experimental Design

Quasi-Experiments: IV

Quasi-Experiments: DiD

Conclusion

Causality is something that we are interested in almost every day

Did this new policy actually work?

Does working from home affect productivity?

Does studying at SSE Riga increase your future income?

Answering these professional questions requires more than just observing correlations.

Does studying at SSE Riga increase your future income?

Students who graduate from SSE Riga earn significantly more than the average graduate in the Baltics.

Is it the education?
The rigorous curriculum, network, and brand value cause the higher salary.

Or is it the students?
The school admits students who would have earned high salaries anyway.

The Rubin Causal Model formalizes this question

Let $D_i = 1$ if student i attends SSE Riga, and $D_i = 0$ if they do not.

Y_{1i}

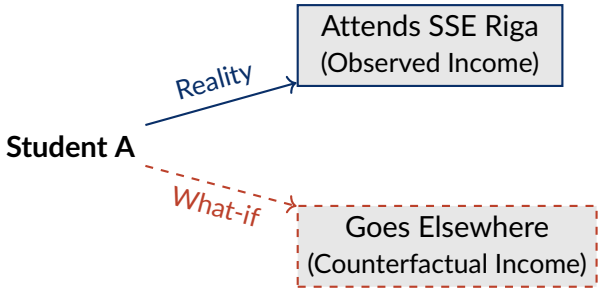
Income
if they attend

Y_{0i}

Income
if they do NOT attend

Individual Treatment Effect $= Y_{1i} - Y_{0i}$

We must find the Counterfactual



Counterfactual: What *would have happened* to an individual if they had been exposed to the other treatment state.
Causal inference is fundamentally about estimating this missing counterfactual data.

Mapping the Notation: Observables vs. Counterfactuals

Treated Units ($D_i = 1$)

$Y_{1i}|D_i = 1$ **(Observed)**
The treated outcome for treated units.

$Y_{0i}|D_i = 1$ **(Counterfactual)**
The untreated outcome for treated units.

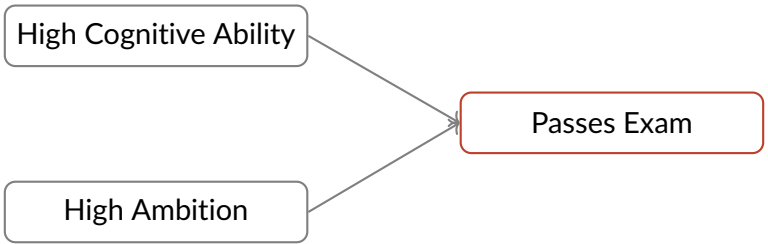
Control Units ($D_i = 0$)

$Y_{0i}|D_i = 0$ **(Observed)**
The untreated outcome for untreated units.

$Y_{1i}|D_i = 0$ **(Counterfactual)**
The treated outcome for untreated units.

The admissions exam violates our ability to compare groups

We want to compare the average income of graduates ($D_i = 1$) to non-graduates ($D_i = 0$).



The traits that cause a student to be admitted are the exact same traits that cause a high salary in the labor market.

Naive comparisons capture both effect and selection bias

When we observe the population, we compute the difference in observed means:

$$E[Y_i|D_i=1] - E[Y_i|D_i=0] = \underbrace{E[Y_{1i}|D_i=1] - E[Y_{0i}|D_i=1]}_{\text{ATT}} + \underbrace{E[Y_{0i}|D_i=1] - E[Y_{0i}|D_i=0]}_{\text{Selection Bias}}$$

Because admitted students have higher ability, their average **counterfactual** income is higher: $E[Y_{0i}|D_i=1] > E[Y_{0i}|D_i=0]$. The naive gap is inflated.

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Selection bias in regression: Omitted Variable Bias

In empirical work, economists estimate relationships using linear regression (OLS):

$$Y_i = \alpha_0 + \alpha_1 D_i + u_i$$

With a binary treatment $D_i \in \{0, 1\}$, the OLS slope equals the difference in sample means:

$$\hat{\alpha}_1 \xrightarrow{p} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$$

- What is packed inside the error term u_i?
  - Innate ability, ambition, work ethic, family background...
- When admitted students possess higher ability:

$$\text{Cov}(D_i, u_i) \neq 0 \implies E[u_i \mid D_i] \neq 0$$

- The zero conditional mean assumption is violated. **Selection bias** in potential outcomes is the exact econometric equivalent of **Omitted Variable Bias** in OLS.

The Omitted Variable Bias Formula

Suppose the true causal model includes ability A_i:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 A_i + \varepsilon_i \quad \text{with } E[\varepsilon_i \mid D_i, A_i] = 0$$

where β_1 is the **true causal effect** and β_2 is the effect of ability on earnings.

If ability is omitted and we run the "short" regression $Y_i = \alpha_0 + \alpha_1 D_i + u_i$, the estimated coefficient converges to:

$$\alpha_1 = \beta_1 + \underbrace{\beta_2 \times \delta_1}_{\text{Omitted Variable Bias}}$$

where $\delta_1 = \frac{\text{Cov}(D_i, A_i)}{\text{Var}(D_i)}$ is the slope from the auxiliary regression of A_i on D_i:

$$\text{Bias} = \underbrace{\beta_2}_{\substack{\text{Impact of Omitted} \\ \text{Variable on } Y}} \times \underbrace{\frac{\text{Cov}(D_i, A_i)}{\text{Var}(D_i)}}_{\substack{\text{Relationship between} \\ \text{Treatment and Omitted Variable}}}$$

Signing the Bias: The Intuition

The sign of the bias is determined by the product of two relationships:

$$\text{sign}(\text{Bias}) = \text{sign}(\beta_2) \times \text{sign}(\text{Cov}(D_i, A_i))$$

	$\text{Cov}(D, A) > 0$	$\text{Cov}(D, A) < 0$
$\beta_2 > 0$	**Positive Bias** (Overestimate)	**Negative Bias** (Underestimate)
$\beta_2 < 0$	**Negative Bias** (Underestimate)	**Positive Bias** (Overestimate)

Applying to SSE Riga:

- $\beta_2 > 0$: Ability increases future earnings.
- $\text{Cov}(D, A) > 0$: Higher-ability individuals are more likely to attend SSE Riga ($\delta_1 > 0$).
- $\implies$ **Positive Bias**: Naive OLS overstates the causal return to attending SSE Riga!

Can we just control for ability?

- **Selection on Observables**: If we could observe and control for ability:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 A_i + \varepsilon_i$$

  Conditioning on A_i eliminates the bias ($\hat{\beta}_1$ identifies the true causal effect).

- **Why this fails in practice**:
  1. **Unobservability**: Key confounders (drive, grit, cognitive ability) cannot be perfectly measured.
  2. **Imperfect Proxies**: Using SAT scores or high school GPA leaves residual unobserved ability in the error term.
  3. **Bad Controls**: Controlling for variables that are themselves affected by treatment (e.g. post-graduation job title) introduces new bias.

When we cannot control for every confounder, how do we establish causality?

How might we get at the true answer?

If we cannot simply compare the treated and the untreated, what do we do?

We must find a way to eliminate the selection bias term.
We need a control group whose observed outcomes perfectly mirror the **counterfactual** outcomes of the treated group:
$E[Y_{0i}|D_i = 1] = E[Y_{0i}|D_i = 0]$.

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Randomized Controlled Trials eliminate selection bias

If we flip a coin to determine **treatment assignment** ($Z_i \in \{0, 1\}$), assignment is completely unrelated to underlying ability:

$$(Y_{1i}, Y_{0i}) \perp Z_i$$

With **perfect compliance**, actual treatment received equals assignment ($D_i = Z_i$). Both groups are statistically identical:

$$E[Y_{0i} \mid D_i = 1] = E[Y_{0i} \mid D_i = 0]$$

Selection bias vanishes.

The simple difference in observed means between treated and control is exactly the causal effect (ATE):

$$\begin{aligned} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] \\ = E[Y_{1i} - Y_{0i}] \end{aligned}$$

Imperfect compliance requires rescaling the effect

What if we randomize assignment ($Z_i = 1$), but some people refuse treatment ($D_i = 0$)?

The Intent-to-Treat (ITT)
The effect of the coin flip on the outcome.
$$E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]$$

To find the effect of the *treatment itself* on those who comply, we rescale the ITT by the compliance rate (the first stage):

$$\text{LATE} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]}$$

Randomized Controlled Trials are not always possible

We cannot randomly assign students to attend or be rejected from SSE Riga.

It is unethical, politically infeasible, and practically impossible. We have to find another way to measure the effect using observational data.

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Quasi-experiments mimic randomization in the real world

When we cannot run an experiment, we look for natural phenomena or policy rules that act *as if* they randomly assigned treatment.

Instrumental Variables (IV)

Differences-in-Differences (DiD)

The Vietnam Draft Lottery (Angrist, 1990)

Does serving in the military reduce your future earnings?

Men who volunteer for the military might have different civilian earnings potential than those who don't (Selection Bias).

The Natural Experiment
In 1969, the US government used a random lottery (based on birth dates) to assign draft sequence numbers, determining **draft eligibility**.

The lottery is just an encouragement

Draft eligibility from the lottery (Z_i) did not perfectly force military service (D_i).

Draft Eligible ($Z_i = 1$)
(Low lottery number called)

Many were drafted into service ($D_i = 1$).
Never-takers: Some failed the physical exam, enrolled in college (deferment), or avoided service ($D_i = 0$).

Draft Ineligible ($Z_i = 0$)
(High lottery number not called)

Most did not serve in the military ($D_i = 0$).
Always-takers: Some volunteered to enlist anyway ($D_i = 1$).

This is an **imperfect compliance** problem: Z_i is an instrument (encouragement), not treatment receipt! We use LATE (Wald estimator) to recover the causal effect of service on earnings for **compliers**.

Instrumental Variables rely on three core assumptions

For draft eligibility (Z_i) to be a valid instrument for military service (D_i):

1. Relevance

Draft eligibility must actually shift the probability of serving. $\text{Cov}(Z_i, D_i) \neq 0$

2. Exclusion

Draft eligibility only affects earnings *through* military service, not directly (birth dates don't affect wages).

3. Monotonicity

Eligibility weakly increases the likelihood of serving for everyone (no "defiers").

The Econometric Setup of IV (2SLS)

We estimate the effect in two stages using Two-Stage Least Squares (2SLS):

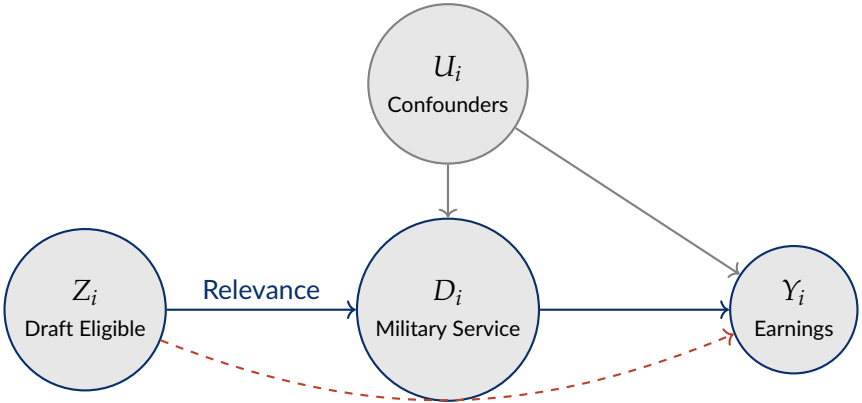
First Stage:

$$\underbrace{D_i}_{\text{Military Service}} = \underbrace{\gamma_0}_{\text{Constant}} + \underbrace{\gamma_1}_{\text{First-Stage Effect}} \underbrace{Z_i}_{\text{Draft Eligible}} + \underbrace{\nu_i}_{\text{Error}}$$

Second Stage:

$$\underbrace{Y_i}_{\text{Earnings}} = \underbrace{\beta_0}_{\text{Constant}} + \underbrace{\beta_1}_{\text{Causal Effect (LATE)}} \underbrace{\hat{D}_i}_{\text{Predicted Service}} + \underbrace{\varepsilon_i}_{\text{Error}}$$

Visualizing the Exclusion Restriction



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Differences-in-Differences: A policy change in the Baltics

Suppose Estonia passes a law allowing citizens to withdraw their pension funds early to spend on current consumption. Latvia does not.

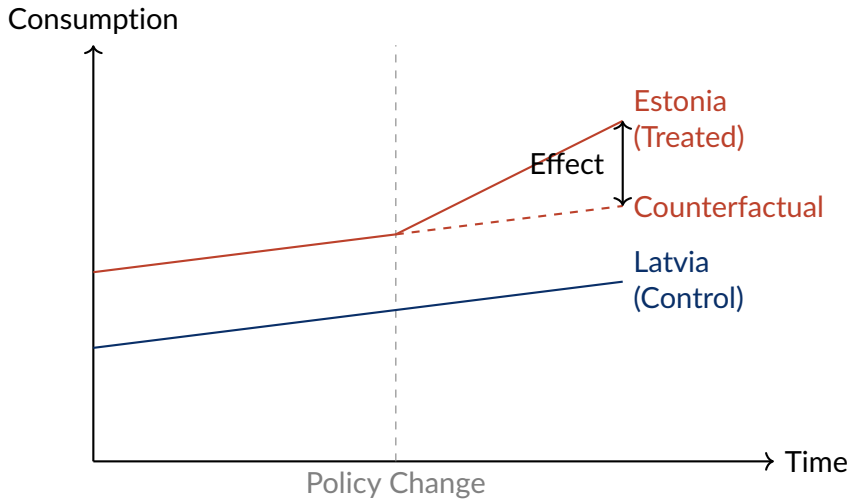
The Question
Does legalizing pension withdrawal increase total household consumption?

The core assumption: Parallel Trends

Estonians and Latvians likely have different baseline consumption levels.

Parallel Trends Assumption
In the absence of the policy change, the **counterfactual** trajectory of consumption in Estonia would have evolved exactly parallel to the trajectory in Latvia.

Differences-in-Differences visually

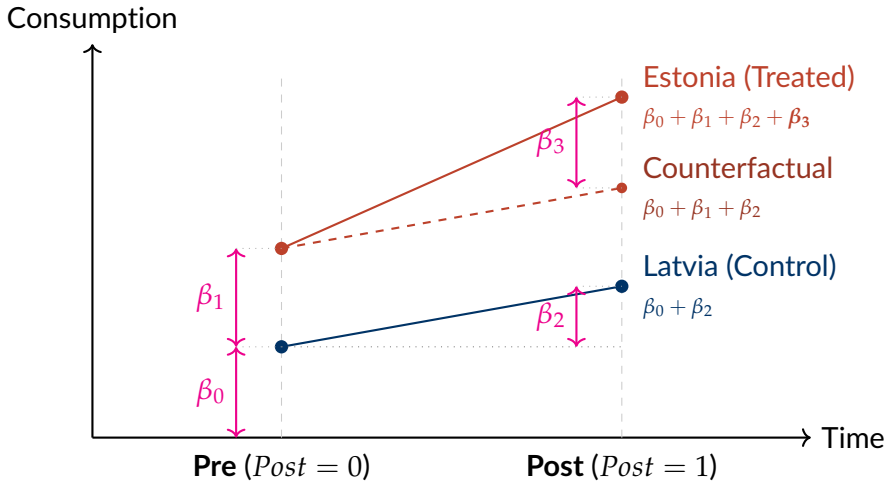


The Econometric Setup of DiD

We estimate the effect using a regression with interaction terms:

$$\underbrace{Y_{it}}_{\text{Consumption}} = \underbrace{\beta_0}_{\text{Latvia Baseline}} + \underbrace{\beta_1 Treat_i}_{\text{Estonia Diff}} + \underbrace{\beta_2 Post_t}_{\text{Time Trend}} + \underbrace{\beta_3 (Treat_i \times Post_t)}_{\text{Causal Effect}} + \underbrace{\varepsilon_{it}}_{\text{Error}}$$

Mapping the Equation to the Graph



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Bringing it all together

Causal inference is the search for the missing **counterfactual**.

Selection bias in potential outcomes is **omitted variable bias** in regression.

RCTs flip a coin to eliminate selection bias by design.

IV uses an exogenous shock/lottery to isolate clean variation in treatment.

DiD uses parallel trends to construct a counterfactual over time.

Assignment: Causal Inference in the Wild

The Task

Find a news article making a causal claim.

The Breakdown

1. Define the Treatment (D_i) and Outcome (Y_i).
2. Explain the unobserved **counterfactual** for the treated group.
3. Identify the source of **selection bias / omitted variables** that inflates or deflates the naive comparison. Sign the bias.
4. What additional **data** would you need to be able to get a causal estimate? Which **population** would this apply to?